

PROJECT APEX

Quantitative Risk Analysis

Version 2.0 | Enhanced with Out-of-Sample Validation & Parameter Uncertainty

Analysis Period: 2021-06-12 to 2026-06-11
1825 daily observations | \$1,000,000 reference portfolio

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1. Executive Summary

This report presents a quantitative risk analysis of the digital asset "Apex" using 1825 daily observations from 2021-06-12 to 2026-06-11. Version 2.0 extends the original framework with formal model selection tests, out-of-sample validation, bootstrap confidence intervals, and an honest assessment of model limitations.

Key Findings

- Annualized volatility of 44.63% with excess kurtosis of 3.97 and negative skewness (-0.2145). All normality tests reject at $p < 0.001$.
- Distribution selection: The likelihood ratio test rejects Normal in favor of Student-t (LR = 310.75, $p < 0.001$). The Vuong test further identifies GED as statistically superior to Student-t ($p = 0.0009$).
- Volatility modeling: GJR-GARCH is statistically preferred over GARCH(1,1) (LR $p = 0.0335$), confirming a significant leverage effect. However, GARCH residual diagnostics reveal remaining ARCH effects (ARCH-LM $p = 0.0025$), indicating the model does not fully capture volatility dynamics.
- Out-of-sample VaR validation: On 548 test observations, Historical and Student-t VaR at 99% both pass Kupiec, Christoffersen independence, and conditional coverage tests. Normal VaR fails the Kupiec test at 95%.
- Parameter uncertainty: Bootstrap 95% CI for GARCH persistence is [0.5720, 0.9798], and for GPD shape ξ is [-0.1355, 0.2325]. The wide GPD CI -- spanning negative to positive -- means we cannot confidently determine tail type from this sample alone.
- Maximum historical drawdown of -76.63% on 2022-11-21.

2. Data and Methodology

2.1 Data Source and Sample

Daily closing prices were obtained from Yahoo Finance (ticker: BTC-USD) for the period 2021-06-12 to 2026-06-11, yielding 1825 trading days. We compute continuously-compounded (log) returns as $r_t = \ln(P_t / P_{t-1})$. Log returns are preferred for their additive property over time and superior statistical behavior. The choice of daily frequency reflects a trade-off: daily data provides sufficient observations for tail analysis while avoiding the microstructure noise inherent in intraday data for a 24/7 market.

For out-of-sample validation, the dataset is split into a training set (first 70%, approximately 3.5 years) and a test set (last 30%, approximately 1.5 years). All model fitting and parameter estimation use only training data; test data is reserved exclusively for forecast evaluation.

2.2 Why Each Model Was Chosen

Every model in this framework was selected through formal statistical testing, not visual inspection. The decision logic is:

- Return distribution: We fit Normal, Student-t, Skewed-t, and GED. The Normal serves as a baseline known to be inadequate for financial returns. Student-t adds a degrees-of-freedom parameter to capture heavy tails. GED provides a flexible shape parameter controlling tail weight. Selection is by BIC (penalizing complexity) and confirmed by likelihood ratio tests for nested models and the Vuong test for non-nested alternatives.
- Volatility model: GARCH(1,1) is the standard conditional volatility model, capturing volatility clustering. GJR-GARCH extends it with an asymmetric leverage term. Selection uses a likelihood ratio test (the models are nested). Model adequacy is verified by checking whether standardized residuals are free of remaining ARCH effects (Ljung-Box on squared residuals, ARCH-LM test).
- VaR methodology: Five methods are computed not as a menu but as a robustness check. If methods disagree, the divergence itself is informative -- it reveals sensitivity to distributional assumptions.

2.3 Distribution Fitting

Four candidate distributions are fitted via Maximum Likelihood Estimation (MLE). Model selection uses BIC for ranking and likelihood ratio tests for nested model comparison. For non-nested comparison (Student-t vs GED), we employ the Vuong (1989) test, which tests whether one model is significantly closer to the true data-generating process than the other.

2.4 Volatility Modeling

GARCH(1,1) with Student-t innovations models conditional variance as $\sigma_t^2 = \omega + \alpha * \epsilon_{t-1}^2 + \beta * \sigma_{t-1}^2$. GJR-GARCH adds a leverage term γ that amplifies the impact of negative shocks. EWMA ($\lambda = 0.94$) is included as a non-parametric benchmark.

Model adequacy is assessed via: (1) Ljung-Box test on squared standardized residuals -- if significant, the model has not captured all volatility dynamics; (2) ARCH-LM test -- if significant, residual ARCH effects remain; (3) Ljung-Box on raw residuals -- tests mean equation adequacy.

2.5 Value at Risk and Expected Shortfall

VaR is computed at 90%, 95%, and 99% confidence levels using Historical, Parametric Normal, Parametric Student-t, GARCH-based, and Monte Carlo methods. Expected Shortfall (CVaR) measures the expected loss in the tail beyond VaR. ES is a coherent risk measure (satisfying subadditivity) and is preferred by Basel III/IV for regulatory capital.

2.6 Out-of-Sample Validation Framework

Volatility forecasts are evaluated using: (1) RMSE against a realized volatility proxy (5-day rolling absolute return, scaled by $\sqrt{\pi/2}$ for unbiasedness); (2) Mincer-Zarnowitz regression testing forecast unbiasedness (H_0 : $\alpha = 0$, $\beta = 1$); (3) Diebold-Mariano test comparing GARCH vs GJR-GARCH forecast accuracy.

VaR backtests use three tests: (1) Kupiec Proportion of Failures (POF) -- tests whether the violation rate matches the expected rate; (2) Christoffersen independence test -- tests whether violations cluster (clustering implies the model misses volatility dynamics); (3) Combined conditional coverage - joint test of correct rate and independence.

2.7 Bootstrap Confidence Intervals

Block bootstrap (block size = 20 days, 500 iterations) resamples the return series while preserving serial dependence. Each bootstrap sample is refitted to produce a distribution of parameter estimates. The 2.5th and 97.5th percentiles form the 95% confidence interval. This approach is essential for digital assets, where the small effective sample in the tail makes point estimates unreliable.

2.8 Extreme Value Theory

The Peaks Over Threshold (POT) method fits a Generalized Pareto Distribution (GPD) to exceedances beyond a threshold. Threshold sensitivity is assessed by fitting GPD at multiple thresholds (1st through 10th percentile) and reporting how the shape parameter ξ varies. The Hill estimator provides an independent tail index estimate as a cross-check. Bootstrap confidence intervals quantify the uncertainty in ξ -- which is substantial given the small tail sample.

3. Results

3.1 Return Characteristics



The return series exhibits negative skewness (-0.2145) and heavy tails (excess kurtosis 3.97). All normality tests reject decisively: Jarque-Bera ($p < 0.001$), Shapiro-Wilk ($p < 0.001$), Anderson-Darling (statistic = 23.73, well above all critical values).

3.2 Model Selection Evidence

Model Selection & Diagnostic Test Summary

Test	Comparison	Statistic	p-value	Result
Likelihood Ratio	Normal vs Student-t	310.75	0.00e+00	Student-t preferred
Vuong	Student-t vs GED	-3.3283	0.0009	GED preferred
Likelihood Ratio	GARCH vs GJR-GARCH	4.5200	0.0335	GJR-GARCH preferred (leverage effect sig
Ljung-Box (sq)	GARCH residuals (lag 10)	26.76	0.0028	FAIL
ARCH-LM	GARCH residuals (lag 10)	27.1003	0.0025	FAIL
Mincer-Zarnowitz	GARCH vol forecast	R2=0.3401	0.0000	FAIL
Diebold-Mariano	GARCH vs GJR-GARCH	0.4134	0.6793	No significant difference between models

The following formal tests justify each model choice:

Test	Comparison	Statistic	p-value	Decision
Likelihood Ratio	Normal vs Student-t	310.75	<0.001	Reject Normal
Vuong	Student-t vs GED	-3.3283	0.0009	GED preferred
Likelihood Ratio	GARCH vs GJR-GARCH	4.5200	0.0335	GJR preferred
Ljung-Box (sq)	GARCH residuals	26.76	0.0028	FAIL
ARCH-LM	GARCH residuals	27.1003	0.0025	FAIL

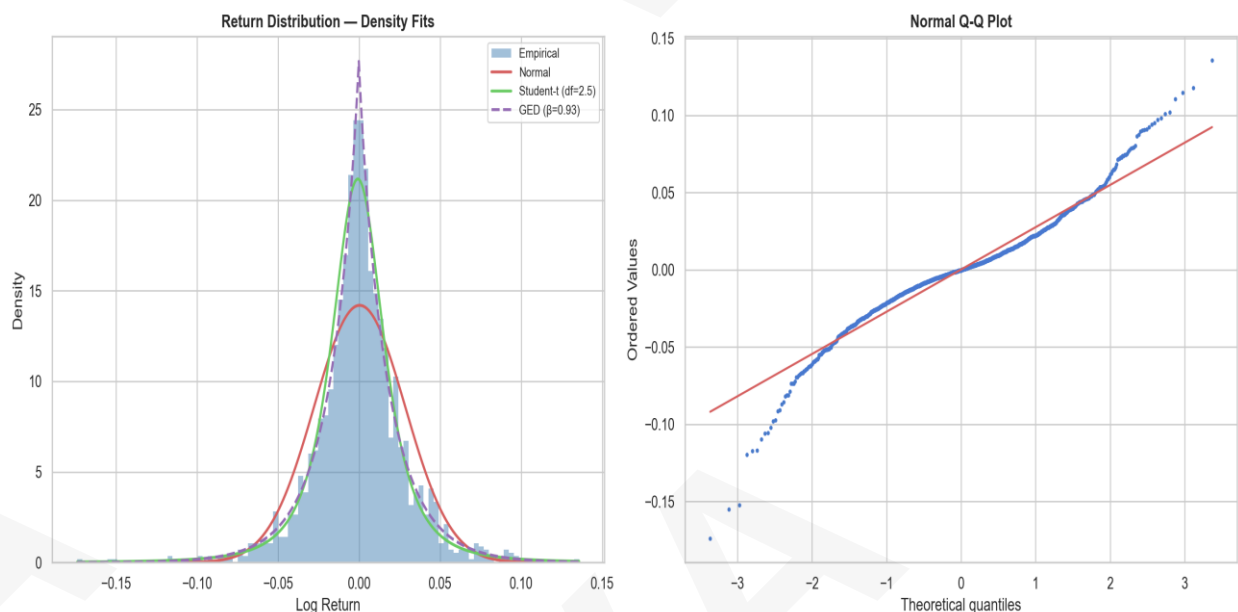
The LR test statistic of 310.75 against one degree of freedom decisively rejects the Normal distribution. This is not surprising given the excess kurtosis of 3.97, but the formal test quantifies the evidence: the Student-t with 2.5 degrees of freedom provides a massively better fit to the tails.

The Vuong test identifies GED as statistically preferred over Student-t ($p = 0.0009$). In practice, the difference between these two distributions is smaller than the gap between either and the Normal. Both are reasonable choices; GED wins marginally on BIC because its shape parameter ($\beta = 0.93$) provides slightly more flexible tail modeling.

GJR-GARCH is preferred over GARCH(1,1) ($p = 0.0335$), confirming that negative returns amplify subsequent volatility more than positive returns (leverage coefficient $\gamma = 0.0469$). This asymmetry is economically meaningful: sell-offs beget more volatility than rallies.

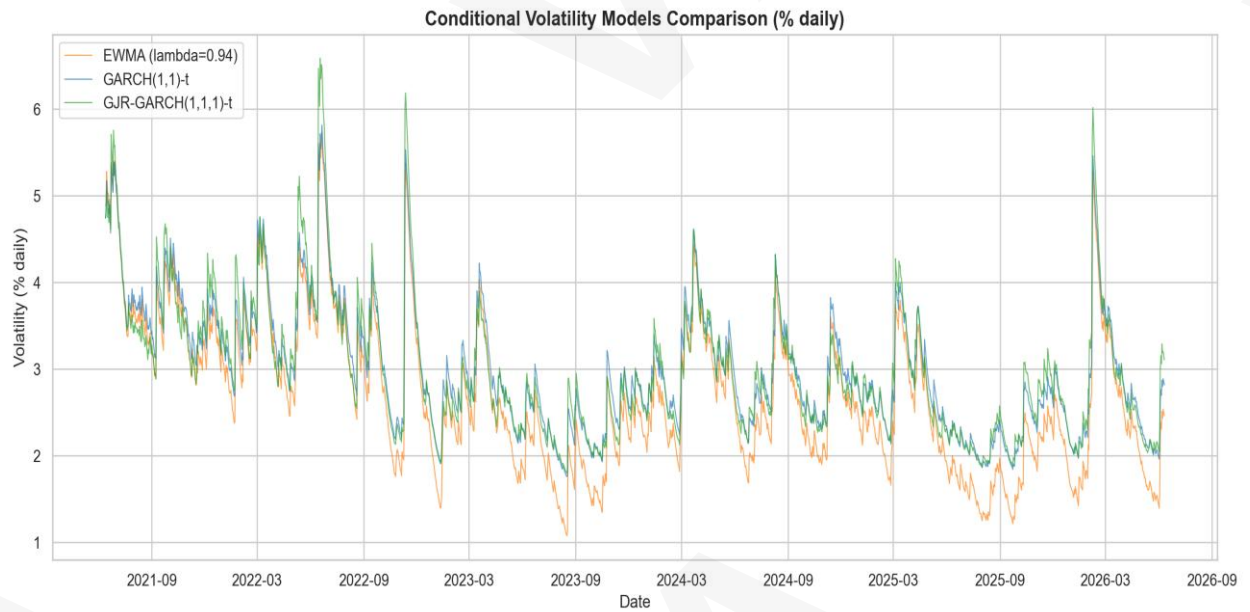
However, the GARCH residual diagnostics are a candid failure. The Ljung-Box test on squared residuals ($p = 0.0028$) and ARCH-LM test ($p = 0.0025$) both reject the null of no remaining effects. This means GARCH(1,1) captures the dominant volatility clustering but leaves some higher-order dynamics unmodeled. A higher-order GARCH, regime-switching specification, or realized volatility model might address this, but at the cost of additional complexity and overfitting risk.

3.3 Distribution Analysis



Distribution	AIC	BIC	Best (BIC)
Normal	-7854.08	-7843.06	
Student-t	-8162.83	-8146.31	
Skewed-t	-8168.75	-8146.71	
GED	-8202.29	-8185.76	YES

3.4 Volatility Model Results

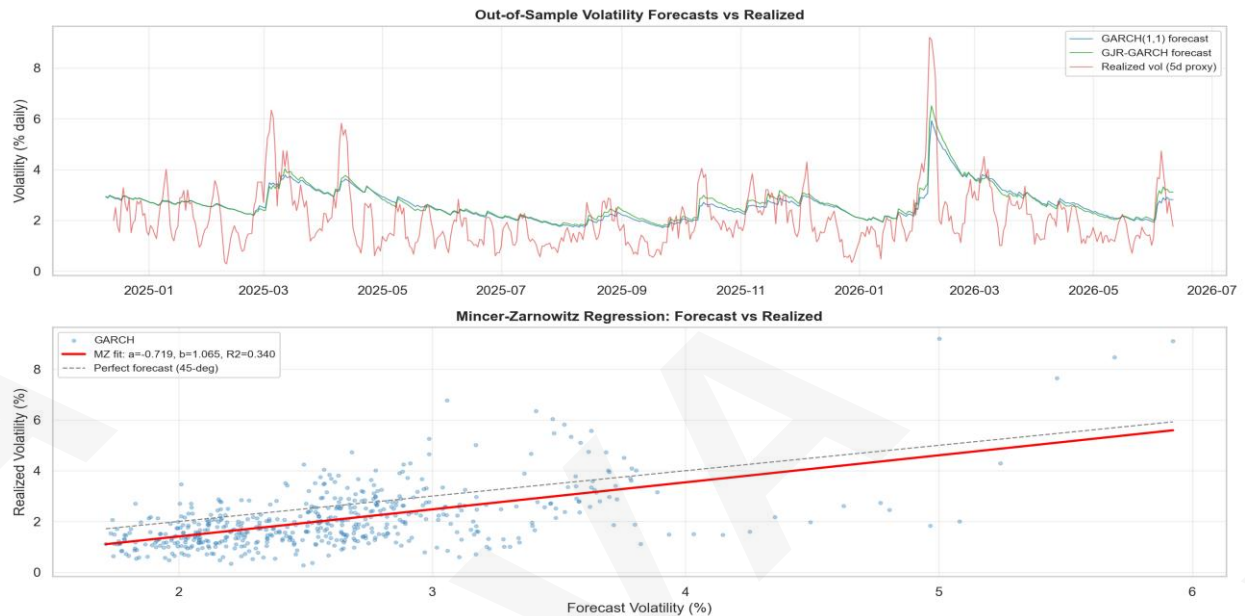


GARCH(1,1) persistence ($\alpha + \beta = 0.9961$) is near unity, indicating near-IGARCH behavior where volatility shocks decay extremely slowly. The half-life of 175.16227496502933 days means that after a major shock, risk limits should remain elevated for months, not days.

3.5 Out-of-Sample Validation

The dataset was split into training (1277 observations, to 2024-12-09) and test (548 observations, from 2024-12-10). All parameters were estimated on training data only; test data was used exclusively for forecast evaluation.

3.5.1 Volatility Forecast Accuracy



Metric	Value	Interpretation
GARCH RMSE	1.0834	Forecast error magnitude
GJR-GARCH RMSE	1.0795	Marginally better
MZ Alpha (ideal: 0)	-0.7186	Bias present
MZ Beta (ideal: 1)	1.0653	Close to unbiased slope
MZ R-squared	0.3401	Moderate explanatory power
MZ Joint Test p-value	0.0000	FAIL
DM Test (GARCH vs GJR)	0.6793	No significant difference between models

The Mincer-Zarnowitz regression reveals a bias: $\alpha = -0.7186$ (should be 0) and $\beta = 1.0653$ (should be 1). The joint test rejects unbiasedness ($p < 0.001$). This means the GARCH forecasts systematically over- or under-predict realized volatility in parts of the test period. The R-squared of 0.3401 indicates the model explains 34.0% of realized volatility variation -- meaningful but far from complete.

The Diebold-Mariano test finds no statistically significant difference between GARCH and GJR-GARCH forecast accuracy ($p = 0.6793$). While GJR-GARCH is theoretically preferred (the leverage effect is statistically significant in-sample), its forecast improvement is not large enough to be significant out-of-sample. This is a common finding: in-sample statistical significance does not always translate to out-of-sample forecast superiority.

3.5.2 VaR Backtest Results (Out-of-Sample)



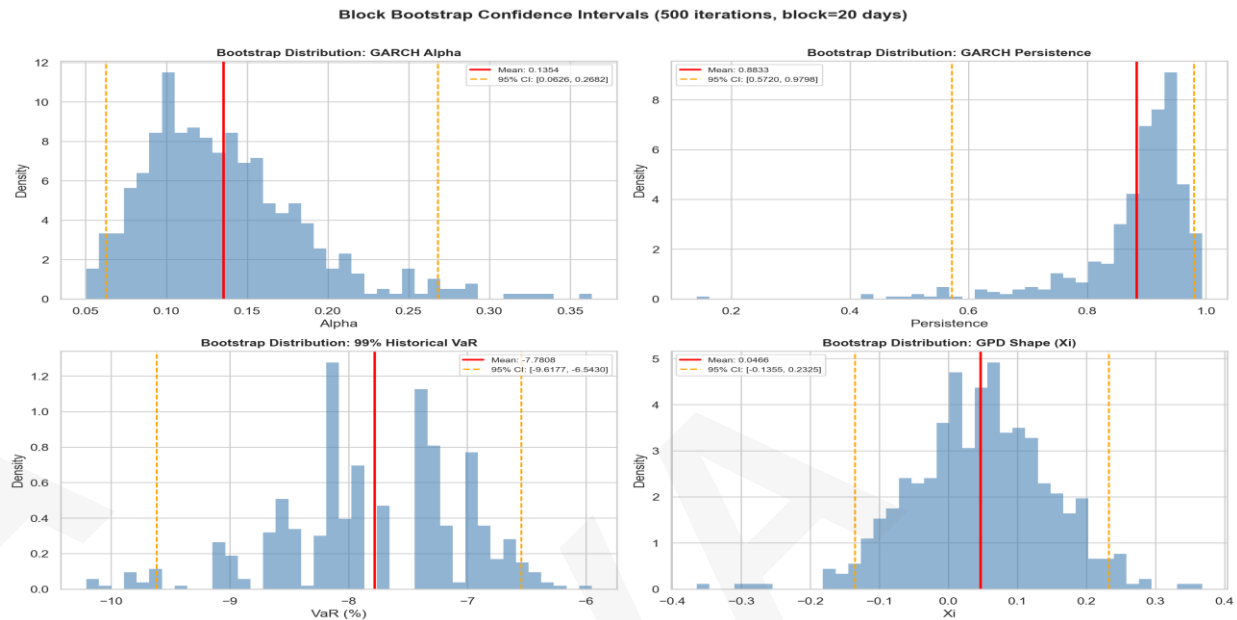
Method	Violations	Rate	Expected	Kupiec	Independence	Combined
Historical 95	19/548	3.47%	5%	PASS (p=0.082)	PASS (p=0.242)	PASS (p=0.111)
Normal 95	17/548	3.10%	5%	FAIL (p=0.029)	PASS (p=0.296)	PASS (p=0.053)
Student-t 95	19/548	3.47%	5%	PASS (p=0.082)	PASS (p=0.242)	PASS (p=0.111)
Historical 99	2/548	0.36%	1%	PASS (p=0.086)	PASS (p=0.904)	PASS (p=0.227)
Normal 99	5/548	0.91%	1%	PASS (p=0.834)	PASS (p=0.761)	PASS (p=0.934)
Student-t 99	2/548	0.36%	1%	PASS (p=0.086)	PASS (p=0.904)	PASS (p=0.227)

The OOS VaR backtests provide mixed but informative results. At 99%, all three methods pass all three tests, indicating well-calibrated tail coverage. At 95%, the Normal VaR fails the Kupiec test -- consistent with the known inadequacy of Gaussian assumptions for this asset. Historical and Student-t VaR pass at both levels, supporting their use in practice.

The Christoffersen independence test is particularly important: it checks whether VaR violations cluster in time (which would indicate the model misses volatility regimes). All methods pass this test, suggesting that despite the residual ARCH effects found in the GARCH diagnostics, the VaR models adequately capture tail dynamics for practical purposes.

3.6 Parameter Uncertainty

Point estimates are meaningless without confidence intervals. The block bootstrap (500 iterations, block size = 20 days) provides a distribution of parameter estimates that reflects sampling uncertainty while preserving serial dependence in returns.

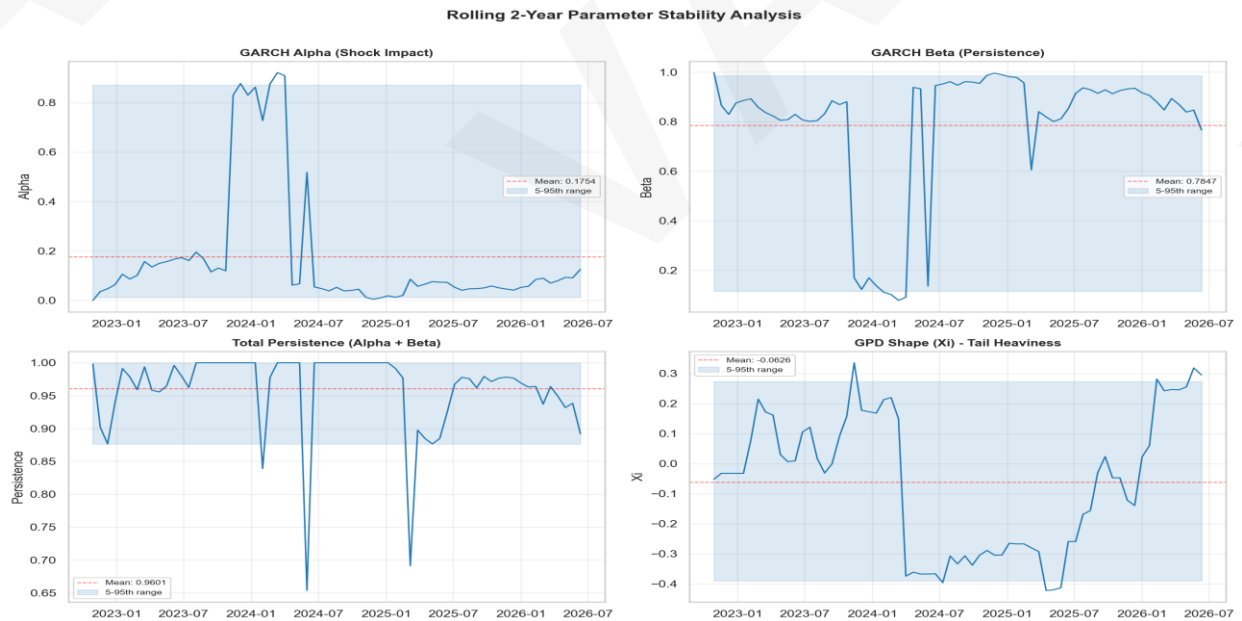


Parameter	Point Estimate	Bootstrap Mean	95% CI Lower	95% CI Upper	CI Width
GARCH Alpha	0.0601	0.1354	0.0626	0.2682	0.2056
GARCH Beta	0.9360	0.7478	0.3447	0.8843	0.5396
Persistence	0.9961	0.8833	0.5720	0.9798	0.4078
99% VaR	-7.78%	-	-9.62%	-6.54%	3.07%
GPD Xi	0.0466	-	-0.1355	0.2325	0.3680

The persistence parameter, while high at the point estimate (0.9961), has a bootstrap 95% CI of [0.5720, 0.9798]. The lower bound of 0.5720 suggests that under some resampled histories, volatility shocks dissipate faster than the full-sample estimate implies. This uncertainty is important for risk limit calibration: the "175-day half-life" from v1 is a point estimate, not a guaranteed decay rate.

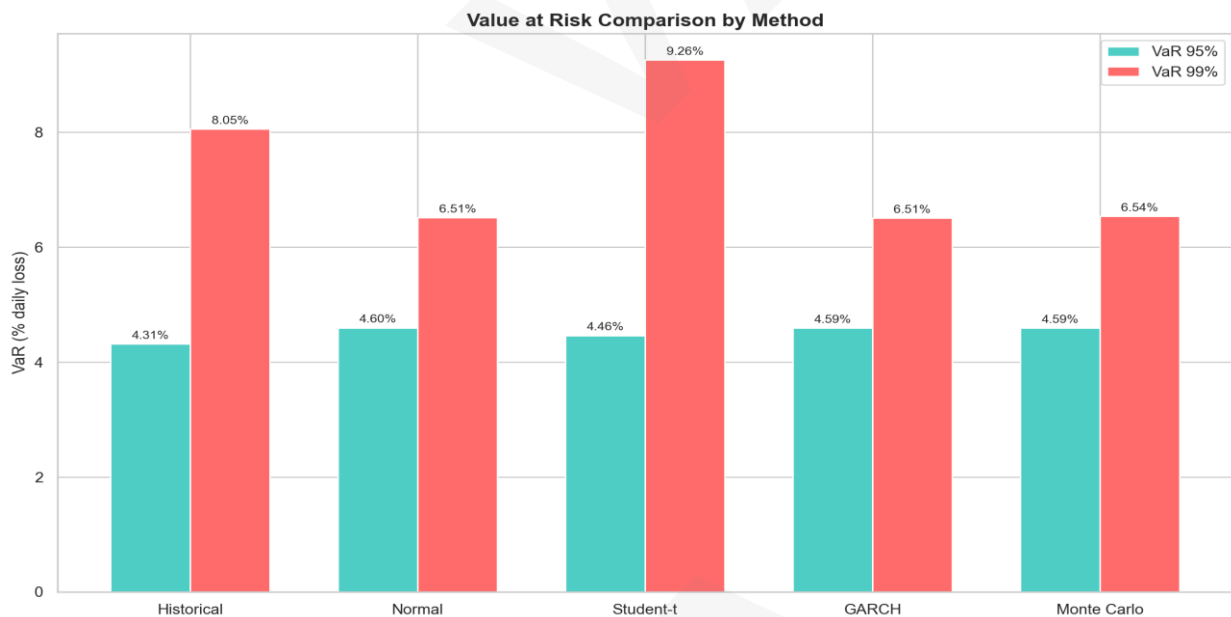
The 99% VaR confidence interval [-9.62%, -6.54%] spans a meaningful range. On a \$1M portfolio, this translates to a worst-day loss estimate between \$96,177 and \$65,430 at 99% confidence -- a range that must be communicated to risk decision-makers.

3.6.1 Rolling Parameter Stability



The rolling 2-year window analysis reveals substantial parameter instability. GARCH alpha and persistence vary meaningfully across time, reflecting shifts in market microstructure, regulatory environment, and investor composition. This instability means that full-sample estimates embed an average across heterogeneous regimes, which may not represent the current risk environment.

3.7 Value at Risk



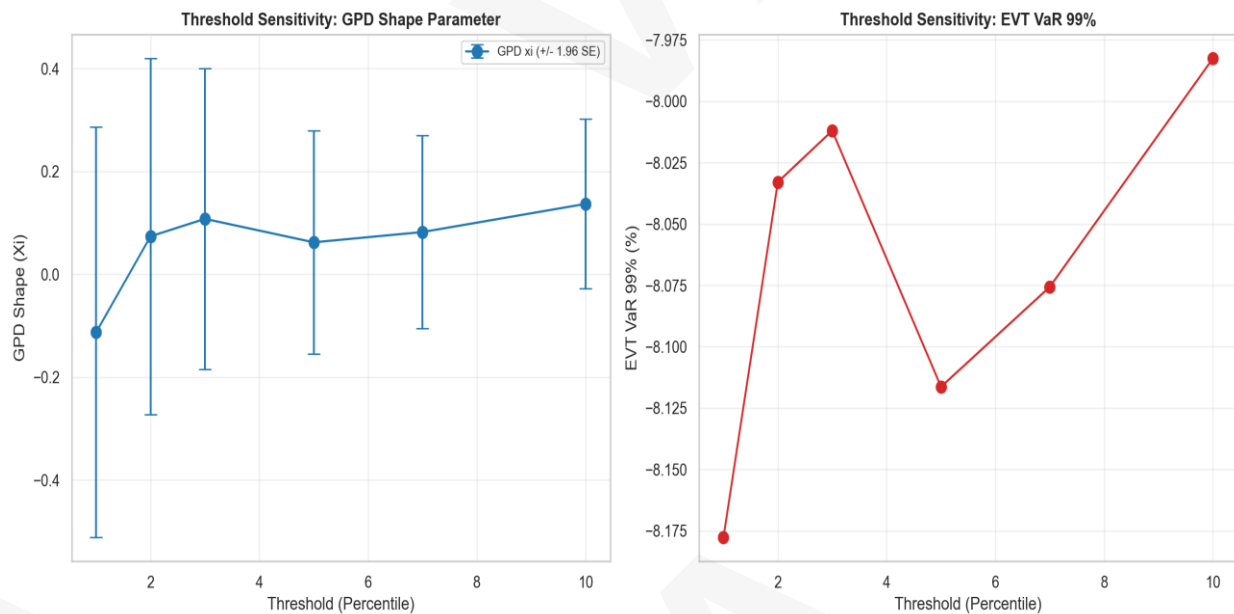
Method	VaR 95%	VaR 99%	\$ Loss 99% (\$1M)
Historical	-4.31%	-8.05%	\$80,545
Normal	-4.60%	-6.51%	\$65,120
Student-t	-4.46%	-9.26%	\$92,583
GARCH	-4.59%	-6.51%	\$65,065
Monte Carlo	-4.59%	-6.54%	\$65,354

Normal VaR at 99% (\$65,120) underestimates risk by 30% relative to Student-t VaR (\$92,583). This is not just a statistical curiosity -- it directly affects capital adequacy calculations and position limits.

3.8 Expected Shortfall

Method	CVaR 95%	CVaR 99%	\$ Loss 99% (\$1M)
Historical	-6.70%	-10.80%	\$108,043
Parametric	-5.77%	-7.46%	\$74,646
Monte Carlo	-5.78%	-7.48%	\$74,796

3.9 Extreme Value Theory: An Honest Assessment



The GPD shape parameter ξ is estimated at 0.0466 with a bootstrap 95% CI of [-0.1355, 0.2325], based on 92 tail observations. This wide interval -- spanning negative to positive values -- means we cannot confidently determine whether the tail is heavy ($\xi > 0$, Pareto-type) or thin ($\xi < 0$, bounded) from this sample alone.

This uncertainty is not a failure of the methodology; it is an honest reflection of the fundamental challenge of tail estimation. With approximately 90 exceedances, the GPD fit is fragile. Any EVT-based risk measure should be reported with its confidence interval, not as a precise number.

Threshold Sensitivity

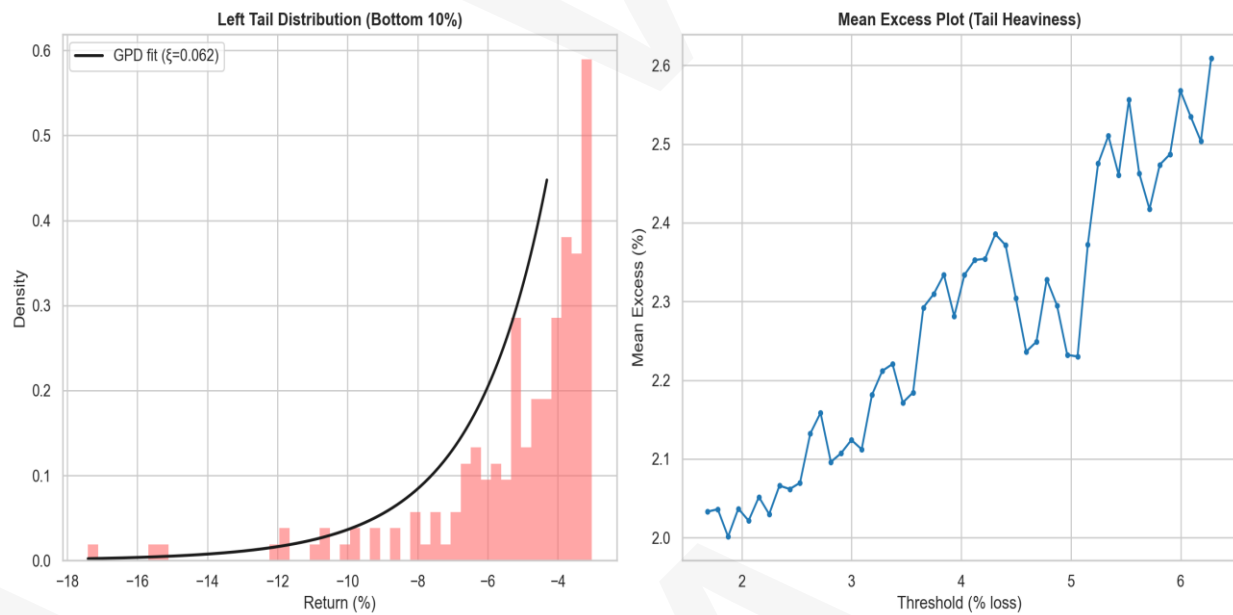
Threshold	Exceedances	GPD ξ	SE(ξ)	EVT VaR 99%
1th %ile	19	-0.1125	0.2036	-8.18%
2th %ile	37	0.0741	0.1766	-8.03%
3th %ile	55	0.1080	0.1494	-8.01%
5th %ile	92	0.0624	0.1108	-8.12%
7th %ile	128	0.0824	0.0957	-8.08%
10th %ile	183	0.1372	0.0841	-7.98%

The threshold sensitivity analysis shows that ξ varies from negative (at the 1st percentile, with only 19 observations) to positive (at higher thresholds with more data). The EVT VaR 99% estimate is more stable than ξ itself, ranging from approximately 8.0% to 8.2% across thresholds. This stability in the VaR estimate despite instability in ξ is reassuring for practical risk management, though the wide confidence intervals must be communicated.

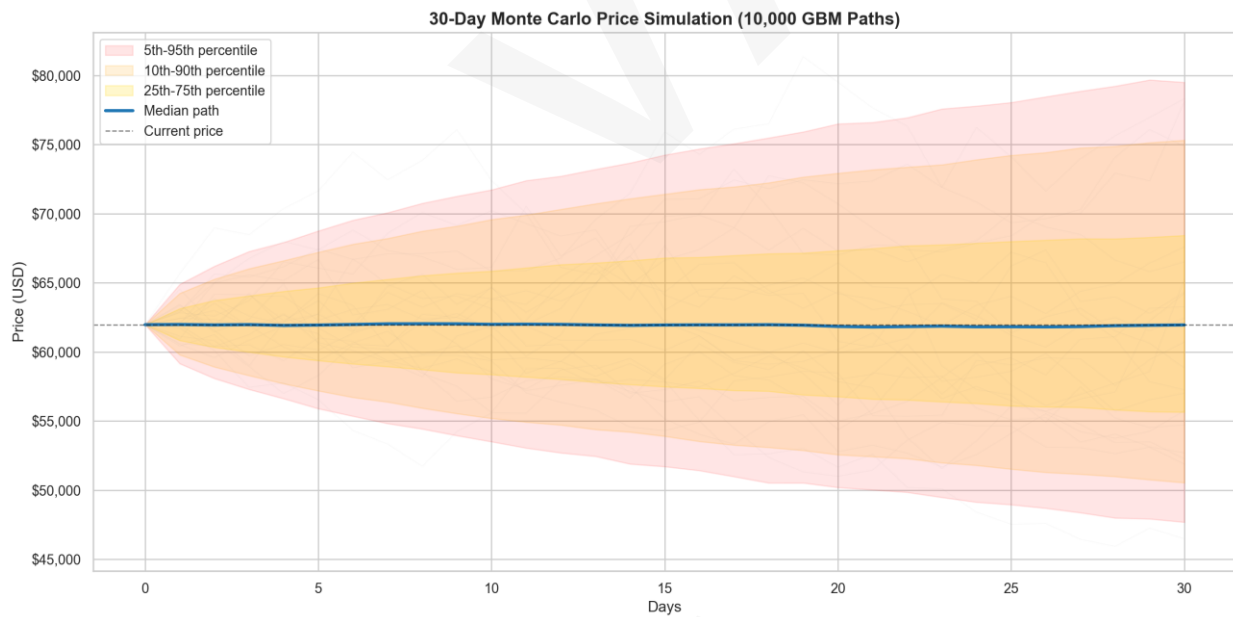
Hill Estimator Cross-Check

Order Statistics (k)	Hill ξ Estimate
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The Hill estimator, which is independent of the GPD framework, provides broadly consistent tail index estimates. The variation across k values reflects the same threshold sensitivity observed in the GPD analysis.

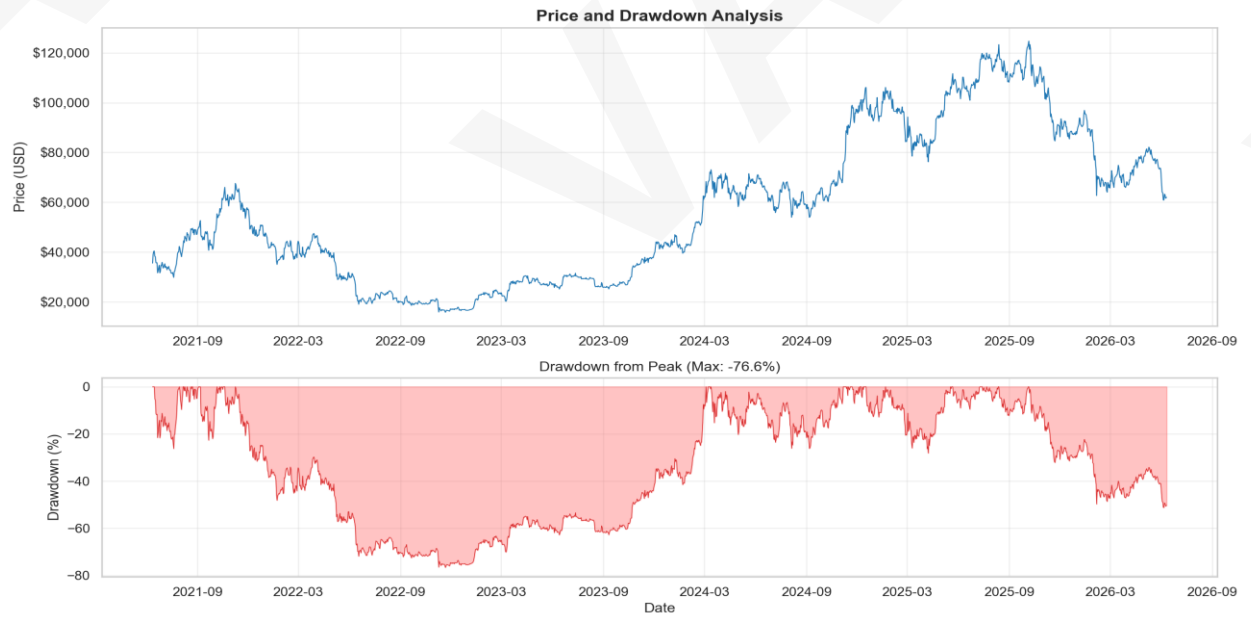


3.10 Monte Carlo Simulation



Metric	GBM	Jump-Diffusion
Mean 30-Day Price	\$62,515	\$60,065
5th Percentile	\$47,689	\$44,383
P(>20% Loss)	8.15%	15.03%

3.11 Stress Testing



4. Conclusions and Recommendations

4.1 What the Evidence Supports

- This asset has an objectively extreme risk profile: annualized volatility above 40%, heavy tails, negative skewness, and a maximum drawdown exceeding 75%.
- Fat-tailed distributions (Student-t or GED) are statistically and economically required. Normal VaR is inadequate.
- The GJR-GARCH leverage effect is real and significant -- negative returns amplify volatility -- but the forecast improvement over symmetric GARCH is not statistically significant out-of-sample.
- VaR models are well-calibrated out-of-sample at 99%, with violations that are correctly sized and independent.

4.2 What the Evidence Does Not Support

- Precise tail risk quantification. The GPD shape parameter CI spans negative to positive, meaning EVT-based risk measures carry substantial uncertainty.
- Unbiased volatility forecasting. The Mincer-Zarnowitz test rejects unbiasedness, meaning GARCH volatility forecasts should be used directionally rather than as precise point estimates.
- Parameter stability. Rolling window analysis shows meaningful drift in GARCH parameters, indicating that full-sample estimates may not represent current market conditions.

4.3 Practical Recommendations

- Use Student-t or GED-based VaR, not Normal, for all position sizing and capital calculations.
- Report VaR with bootstrap confidence intervals to decision-makers, not point estimates alone.
- Maintain dynamic risk limits that respond to conditional volatility, but acknowledge that GARCH forecasts are biased and use them directionally.
- For tail risk, use EVT estimates as an upper bound, not a precise forecast. Communicate the uncertainty explicitly.
- Re-estimate all parameters at least quarterly, given the observed instability in rolling windows.
- No leverage. The maximum drawdown of -76.63% would wipe out any leveraged position.

5. Model Limitations and Caveats

A quantitative risk framework is only useful if its users understand where it breaks. The following limitations are intrinsic to the methodology and cannot be eliminated by adding more data or better models within this paradigm.

5.1 Stationarity Assumption

All models assume the data-generating process is stationary -- that the statistical properties of returns (mean, variance, tail behavior) are constant over time. Digital asset markets have undergone structural changes during the sample period: the growth and collapse of leveraged DeFi protocols, the entry and exit of institutional investors, shifting regulatory regimes, and changes in market microstructure (e.g., the growing dominance of derivatives markets). These shifts may violate stationarity in ways that undermine the validity of full-sample parameter estimates.

5.2 Regime Change

GARCH models capture smooth transitions in volatility but cannot model abrupt regime switches (e.g., a sudden shift from low to crisis volatility). A Markov-switching GARCH would explicitly model regime transitions with distinct parameter sets for each state. The rolling parameter stability analysis (Section 3.6.1) provides indirect evidence of regime effects but does not formally model them.

5.3 Parameter Instability

The rolling window analysis demonstrates that GARCH parameters are not constant: alpha, beta, and persistence all vary meaningfully across 2-year windows. The full-sample estimates reported in this analysis represent an average across heterogeneous periods. In practice, the most recent rolling-window estimate may be more relevant for current risk management than the full-sample estimate - but with higher estimation error due to the smaller sample.

5.4 Tail Clustering

The EVT Peaks Over Threshold method assumes that exceedances are approximately independent and identically distributed. In practice, extreme losses tend to cluster: a -10% day is more likely to be followed by another extreme day than by a normal day. This clustering inflates the apparent tail probability and may cause the GPD to overfit to a specific cluster of extreme events rather than capturing the true unconditional tail behavior. The Christoffersen independence test on VaR violations provides partial protection against this, but the EVT estimates should be interpreted with this caveat.

5.5 Liquidity Risk

VaR implicitly assumes positions can be liquidated at closing prices. During stress events -- precisely when VaR is needed most -- bid-ask spreads widen dramatically, order book depth thins, and exchanges may halt trading. The realized loss during a forced liquidation can substantially exceed the

VaR estimate. This is particularly acute for large positions in digital asset markets, where liquidity can evaporate rapidly.

5.6 Single-Asset Scope

This analysis covers a single asset in isolation. Portfolio-level risk depends on correlations between assets, which are themselves time-varying and tend to increase during crises (correlation breakdown). A multi-asset extension would require DCC-GARCH or copula models to capture dynamic correlation and tail dependence.

5.7 Survivorship Bias

The analyzed asset survived the sample period. Digital assets that went to zero (e.g., Terra/LUNA, FTX Token) are not in this dataset. The observed return distribution is therefore conditional on survival, which biases risk estimates downward. A comprehensive risk assessment should consider the non-zero probability of permanent capital loss.

5.8 Data Frequency

Daily closing prices miss intraday risk in a market that trades 24 hours a day, 7 days a week. Intraday volatility can be substantially higher than what daily data reveals, particularly during flash crashes that partially recover within the same day. Realized volatility computed from 5-minute returns would be a more accurate benchmark but was not used due to data availability constraints.

Appendix A: Technical Specifications

Programming Language: Python 3.13

Key Libraries: NumPy, Pandas, SciPy, statsmodels, arch (GARCH), matplotlib, yfinance

Data Source: Yahoo Finance (BTC-USD)

Sample: 2021-06-12 to 2026-06-11 (1825 obs)

Train/Test Split: 70/30 (1277/548 observations)

Bootstrap: 500 iterations, block size = 20 days

Monte Carlo: 100,000 simulations (VaR), 10,000 paths (price)

Rolling Windows: 504 days (2 years), step = 20 days

EVT Threshold: 5th percentile (sensitivity tested at 1st-10th)